

1 Agroforestry SmartAgroforst: Poplar biomass modelling

2 Thomas Kirschstein^{1,1,1,*}

3 Abstract

Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock option for bio-based value chains, but their economic viability depends on coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains. This paper develops a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with explicit age-lag constraints that link consecutive harvests at each site. The formulation couples allometric yield functions for different biomass types with a multi-product, multi-period network design model including storage, processing, and product quality cascades. The model is applied to a stylised case study of Central European poplar-based alley-cropping systems to explore trade-offs between harvest rotation length, product portfolios, and infrastructure configuration. We outline a computational experimentation plan to test the impact of allometric parameter uncertainty, price scenarios, and policy incentives on the optimal deployment of agroforestry systems in cleaner bioeconomy pathways.

4 *Keywords:* Agroforestry, Short-rotation coppice, Biomass supply chain, Product
5 cascading

```
6 ## [1] "age"          "scenario"      "Merch. stem"  "Merch. branch"  
7 ## [5] "Residue"  
  
8 ## Building sparse LP (Rcpp v10-no-Y, matching !build_AFS_milp.R)...  
9 ##   n_vars=312907 (z:66265 Xij:177045 S:1221 Xjk:68376)  
10 ##   C1: Path establishment
```

*Corresponding author

Email address: thomas.kirschstein@ikts.fraunhofer.de (Thomas Kirschstein)

Preprint submitted to Journal Name

May 29, 2026

```

11  ## C2: Path connectivity
12  ## C3: Biomass yield
13  ## C4: Inventory balance
14  ## C5: Storage capacity
15  ## C6: Processing capacity
16  ## C7: Demand satisfaction
17  ## Total constraints: 27183, nnz: 977322

18  ## [solve_lp_with_gurobi] 312907 vars | 27183 constrs | 977322 nnz | sense=max | start=none
19  ## Set parameter Username
20  ## Set parameter LicenseID to value 2805714
21  ## Set parameter TimeLimit to value 300
22  ## Set parameter MIPGap to value 0.05
23  ## Set parameter MIPFocus to value 1
24  ## Set parameter Cuts to value 3
25  ## Academic license - for non-commercial use only - expires 2027-04-10
26  ## Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 11+.0 (26200.2))
27  ##
28  ## CPU model: AMD Ryzen AI 9 HX 370 w/ Radeon 890M, instruction set [SSE2|AVX|AVX2|AVX512]
29  ## Thread count: 12 physical cores, 24 logical processors, using up to 24 threads
30  ##
31  ## Non-default parameters:
32  ## TimeLimit 300
33  ## MIPGap 0.05
34  ## MIPFocus 1
35  ## Cuts 3
36  ## LogToConsole 0
37  ##
38  ## Optimize a model with 27183 rows, 312907 columns and 977322 nonzeros
39  ## Model fingerprint: 0x88cc9c6c
40  ## Variable types: 246642 continuous, 66265 integer (66265 binary)
41  ## Coefficient statistics:

```

```

42  ## Matrix range      [1e+00, 6e+05]
43  ## Objective range   [1e-01, 2e+07]
44  ## Bounds range      [1e+00, 5e+07]
45  ## RHS range         [1e+00, 7e+07]
46  ## Found heuristic solution: objective -0.0000000
47  ## Presolve removed 3513 rows and 47278 columns
48  ## Presolve time: 2.78s
49  ## Presolved: 23670 rows, 265629 columns, 844989 nonzeros
50  ## Variable types: 204919 continuous, 60710 integer (60710 binary)
51  ## Found heuristic solution: objective 520958.36106
52  ## Deterministic concurrent LP optimizer: primal simplex, dual simplex, and barrier
53  ## Showing barrier log only...
54  ##
55  ## Root barrier log...
56  ##
57  ## Ordering time: 0.42s
58  ##
59  ## Barrier statistics:
60  ## AA' NZ      : 5.969e+05
61  ## Factor NZ   : 7.942e+06 (roughly 200 MB of memory)
62  ## Factor Ops  : 8.060e+09 (less than 1 second per iteration)
63  ## Threads     : 10
64  ##
65  ##              Objective              Residual
66  ## Iter      Primal      Dual      Primal      Dual      Compl      Time
67  ##   0  -3.84038194e+16  8.48936215e+16  2.46e+09  1.05e+06  2.94e+13  4s
68  ##   1  -1.44618807e+16  7.53170607e+16  9.20e+08  1.96e+06  1.13e+13  4s
69  ##   2  -2.34221535e+15  5.73772127e+16  1.42e+08  1.40e+05  1.82e+12  4s
70  ##   3  -5.18549735e+14  3.27112613e+16  2.77e+07  8.87e+03  3.80e+11  4s
71  ##   4  -1.21943699e+14  1.60653485e+16  5.76e+06  2.43e+03  9.21e+10  4s
72  ##   5  -2.23407504e+13  6.40620215e+15  9.00e+05  2.24e+02  2.21e+10  4s

```

73	##	6	-2.72211910e+12	3.51511059e+15	3.50e+04	9.02e+01	7.50e+09	4s
74	##	7	-9.38749178e+11	2.62656276e+15	1.11e+04	5.03e+01	5.09e+09	5s
75	##	8	-4.54594076e+11	9.69737695e+14	4.62e+03	4.06e+00	1.85e+09	5s
76	##	9	-2.34511247e+11	2.16643282e+14	1.67e+03	3.51e-08	4.10e+08	5s
77	##	10	-1.69142677e+11	1.74176730e+13	7.99e+02	3.72e-08	3.57e+07	5s
78	##	11	-1.49034000e+11	1.00040254e+13	5.79e+02	2.33e-08	2.12e+07	5s
79	##	12	-1.22707332e+11	4.69044530e+12	3.30e+02	1.68e-08	1.05e+07	5s
80	##	13	-1.00929313e+11	1.32550187e+12	1.47e+02	1.77e-08	3.47e+06	5s
81	##	14	-8.82134223e+10	9.55131380e+11	6.84e+01	2.24e-08	2.34e+06	5s
82	##	15	-7.20684813e+10	9.17649786e+11	5.82e+01	2.28e-08	2.18e+06	5s
83	##	16	-5.39143341e+10	7.16070064e+11	4.21e+01	2.25e-02	1.68e+06	5s
84	##	17	-5.00621542e+10	5.58720102e+11	3.86e+01	2.23e-02	1.36e+06	6s
85	##	18	-4.31443493e+10	5.30783294e+11	3.25e+01	2.18e-02	1.26e+06	6s
86	##	19	-2.92723727e+10	3.87038667e+11	2.37e+01	1.31e-02	9.01e+05	6s
87	##	20	-2.34158987e+10	3.43724797e+11	1.92e+01	1.12e-02	7.81e+05	6s
88	##	21	-1.52479825e+10	1.60720798e+11	1.32e+01	4.62e-03	3.85e+05	6s
89	##	22	-6.31955567e+09	1.38560256e+11	6.84e+00	3.75e-03	2.95e+05	6s
90	##	23	-3.88789816e+09	8.10910562e+10	5.03e+00	2.09e-03	1.72e+05	6s
91	##	24	-1.95792114e+09	5.74213150e+10	3.57e+00	1.01e-03	1.18e+05	6s
92	##	25	-4.70188951e+08	3.93360167e+10	2.36e+00	5.84e-05	7.71e+04	6s
93	##	26	5.84447447e+07	2.88494595e+10	1.92e+00	8.38e-09	5.53e+04	6s
94	##	27	7.79759064e+08	2.09269492e+10	1.35e+00	6.52e-09	3.82e+04	7s
95	##	28	1.65842849e+09	1.22816273e+10	6.80e-01	4.66e-09	1.97e+04	7s
96	##	29	1.92331752e+09	9.27153863e+09	4.21e-01	3.26e-09	1.35e+04	7s
97	##	30	2.01063775e+09	7.33557005e+09	3.25e-01	3.73e-09	9.78e+03	7s
98	##	31	2.09561320e+09	6.18556736e+09	2.45e-01	2.79e-09	7.49e+03	7s
99	##	32	2.13742475e+09	5.39828864e+09	2.09e-01	1.86e-09	5.96e+03	7s
100	##	33	2.18985044e+09	4.82702158e+09	1.65e-01	2.79e-09	4.82e+03	7s
101	##	34	2.24629651e+09	4.19453389e+09	1.35e-01	2.79e-09	3.55e+03	7s
102	##	35	2.34352781e+09	3.91956724e+09	9.24e-02	2.33e-09	2.87e+03	7s
103	##	36	2.42228566e+09	3.83053249e+09	5.82e-02	1.86e-09	2.57e+03	7s

104	##	37	2.46406638e+09	3.47320800e+09	4.41e-02	2.33e-09	1.84e+03	8s
105	##	38	2.47614745e+09	3.34628852e+09	4.04e-02	1.86e-09	1.59e+03	8s
106	##	39	2.49168673e+09	3.16016434e+09	3.59e-02	1.86e-09	1.22e+03	8s
107	##	40	2.55209277e+09	3.03150892e+09	2.10e-02	1.40e-09	8.74e+02	8s
108	##	41	2.58475169e+09	2.92374020e+09	1.49e-02	9.31e-10	6.18e+02	8s
109	##	42	2.60761610e+09	2.86353167e+09	1.09e-02	1.40e-09	4.66e+02	8s
110	##	43	2.62525487e+09	2.83005178e+09	8.25e-03	1.40e-09	3.73e+02	8s
111	##	44	2.63748129e+09	2.79065978e+09	6.61e-03	9.31e-10	2.79e+02	8s
112	##	45	2.64673170e+09	2.76511470e+09	5.45e-03	2.56e-09	2.16e+02	9s
113	##	46	2.65758010e+09	2.74375958e+09	4.16e-03	2.33e-09	1.57e+02	9s
114	##	47	2.67147336e+09	2.73003344e+09	2.56e-03	9.31e-10	1.07e+02	9s
115	##	48	2.67717423e+09	2.71792173e+09	1.95e-03	9.31e-10	7.43e+01	9s
116	##	49	2.68269813e+09	2.71404502e+09	1.38e-03	1.40e-09	5.71e+01	9s
117	##	50	2.68576181e+09	2.70967561e+09	1.07e-03	9.31e-10	4.36e+01	9s
118	##	51	2.68915227e+09	2.70553154e+09	7.27e-04	9.31e-10	2.99e+01	9s
119	##	52	2.69054514e+09	2.70270561e+09	5.89e-04	1.40e-09	2.22e+01	9s
120	##	53	2.69362069e+09	2.70023979e+09	2.95e-04	1.86e-09	1.21e+01	9s
121	##	54	2.69458454e+09	2.69900501e+09	2.06e-04	9.31e-10	8.06e+00	10s
122	##	55	2.69525577e+09	2.69861820e+09	1.45e-04	9.31e-10	6.13e+00	10s
123	##	56	2.69596893e+09	2.69791848e+09	7.98e-05	9.31e-10	3.55e+00	10s
124	##	57	2.69618461e+09	2.69759495e+09	6.04e-05	9.31e-10	2.57e+00	10s
125	##	58	2.69640556e+09	2.69736820e+09	4.15e-05	9.31e-10	1.75e+00	10s
126	##	59	2.69655329e+09	2.69717956e+09	2.89e-05	9.31e-10	1.14e+00	10s
127	##	60	2.69667091e+09	2.69704758e+09	1.94e-05	9.31e-10	6.87e-01	10s
128	##	61	2.69681208e+09	2.69699579e+09	8.08e-06	9.31e-10	3.35e-01	10s
129	##	62	2.69686647e+09	2.69694883e+09	3.90e-06	9.31e-10	1.50e-01	10s
130	##	63	2.69690599e+09	2.69693828e+09	1.08e-06	1.16e-09	5.88e-02	10s
131	##	64	2.69691304e+09	2.69692526e+09	9.03e-07	1.63e-09	2.23e-02	10s
132	##	65	2.69691721e+09	2.69692169e+09	5.35e-07	1.63e-09	8.16e-03	10s
133	##	66	2.69691943e+09	2.69692002e+09	2.05e-07	2.10e-09	1.08e-03	11s
134	##	67	2.69691963e+09	2.69691978e+09	4.37e-08	2.33e-09	2.68e-04	11s

```

135  ##      68      2.69691969e+09  2.69691970e+09  5.52e-10  1.86e-09  2.17e-06      11s
136  ##      69      2.69691970e+09  2.69691970e+09  8.10e-09  1.43e-09  4.67e-10      11s
137  ##
138  ## Barrier solved model in 69 iterations and 10.91 seconds (14.45 work units)
139  ## Optimal objective 2.69691970e+09
140  ##
141  ##
142  ## Root crossover log...
143  ##
144  ##      19229 DPushes remaining with DInf 0.0000000e+00      11s
145  ##          0 DPushes remaining with DInf 0.0000000e+00      11s
146  ##
147  ##          734 PPushes remaining with PInf 6.2386010e-05      11s
148  ##          0 PPushes remaining with PInf 0.0000000e+00      12s
149  ##
150  ## Push phase complete: Pinf 0.0000000e+00, Dinf 9.9283947e+03      12s
151  ##
152  ##
153  ## Root simplex log...
154  ##
155  ## Iteration      Objective      Primal Inf.      Dual Inf.      Time
156  ##      16700      2.6969197e+09      0.0000000e+00      9.928395e+03      12s
157  ##      16969      2.6969197e+09      0.0000000e+00      0.0000000e+00      12s
158  ## Concurrent spin time: 2.71s (can be avoided by choosing Method=3)
159  ##
160  ## Solved with barrier
161  ##
162  ## Root relaxation: objective 2.696920e+09, 16969 iterations, 11.48 seconds (10.80 work units)
163  ## Total elapsed time = 22.87s (DegenMoves)
164  ## Total elapsed time = 25.85s (DegenMoves)
165  ##

```

166	##	Nodes		Current Node			Objective Bounds			Work	
167	##	Expl	Unexpl	Obj	Depth	IntInf	Incumbent	BestBd	Gap	It/Node	Time
168	##										
169	##	0	0	2.6969e+09	0	1103	520958.361	2.6969e+09	-	-	26s
170	## H	0	0				1.939632e+09	2.6969e+09	39.0%	-	27s
171	## H	0	0				1.942688e+09	2.6969e+09	38.8%	-	27s
172	## H	0	0				1.943291e+09	2.6969e+09	38.8%	-	27s
173	## H	0	0				1.985397e+09	2.6969e+09	35.8%	-	28s
174	## H	0	0				1.999260e+09	2.6969e+09	34.9%	-	28s
175	## H	0	0				2.068574e+09	2.6969e+09	30.4%	-	28s
176	## H	0	0				2.138444e+09	2.6969e+09	26.1%	-	29s
177	## H	0	0				2.142694e+09	2.6969e+09	25.9%	-	29s
178	## H	0	0				2.142999e+09	2.6969e+09	25.8%	-	30s
179	## H	0	0				2.195887e+09	2.6969e+09	22.8%	-	30s
180	## H	0	0				2.201085e+09	2.6969e+09	22.5%	-	30s
181	## H	0	0				2.201964e+09	2.6969e+09	22.5%	-	30s
182	## H	0	0				2.203656e+09	2.6969e+09	22.4%	-	30s
183	## H	0	0				2.208855e+09	2.6969e+09	22.1%	-	30s
184	## H	0	0				2.222924e+09	2.6969e+09	21.3%	-	30s
185	## H	0	0				2.222960e+09	2.6969e+09	21.3%	-	30s
186	## H	0	0				2.225561e+09	2.6969e+09	21.2%	-	30s
187	## H	0	0				2.226317e+09	2.6969e+09	21.1%	-	31s
188	##	0	0	2.6965e+09	0	1228	2.2263e+09	2.6965e+09	21.1%	-	87s
189	## H	0	0				2.340323e+09	2.6965e+09	15.2%	-	120s
190	##	0	0	2.6963e+09	0	1233	2.3403e+09	2.6963e+09	15.2%	-	132s
191	##	0	0	2.6963e+09	0	1237	2.3403e+09	2.6963e+09	15.2%	-	136s
192	##	0	0	2.6963e+09	0	1231	2.3403e+09	2.6963e+09	15.2%	-	136s
193	##	0	0	2.6963e+09	0	1245	2.3403e+09	2.6963e+09	15.2%	-	137s
194	##	0	0	2.6963e+09	0	1245	2.3403e+09	2.6963e+09	15.2%	-	137s
195	##	0	0	2.6959e+09	0	1410	2.3403e+09	2.6959e+09	15.2%	-	195s
196	## H	0	0				2.469006e+09	2.6959e+09	9.19%	-	252s

```

197  ##      0      0 2.6956e+09      0 1438 2.4690e+09 2.6956e+09 9.18%      - 283s
198  ##      0      0 2.6956e+09      0 1436 2.4690e+09 2.6956e+09 9.18%      - 299s
199  ##      0      0      -      0      2.4690e+09 2.6956e+09 9.18%      - 300s
200  ##
201  ## Cutting planes:
202  ## Gomory: 8
203  ## MIR: 184
204  ## Flow cover: 805
205  ##
206  ## Explored 1 nodes (61288 simplex iterations) in 300.03 seconds (492.65 work units)
207  ## Thread count was 24 (of 24 available processors)
208  ##
209  ## Solution count 10: 2.46901e+09 2.46901e+09 2.34032e+09 ... 2.20196e+09
210  ##
211  ## Time limit reached
212  ## Warning: max constraint violation (2.0132e-06) exceeds tolerance
213  ## Warning: max bound violation (2.6103e-06) exceeds tolerance
214  ## Best objective 2.469005829163e+09, best bound 2.695578580530e+09, gap 9.1767%
215  ## [solve_lp_with_gurobi] Status: TIME_LIMIT | ObjVal: 2469005829.1629 | Runtime: 300.5 s

```

216 1. Introduction

217 Agroforestry systems that integrate trees with agricultural crops (silvoarable systems)
 218 are increasingly recognised as a promising land-use option to reconcile climate mitigation,
 219 biodiversity conservation, and rural income generation on the same hectare of land (Toens-
 220 meier, 2017). Temperate agroforestry systems (AFS) based on fast-growing species such
 221 as poplar, willow, alder, and black locust can deliver substantial biomass yields while im-
 222 proving soil quality and providing multiple ecosystem services compared to conventional
 223 monocultures (Huber et al., 2018). Biomasses produced in AFS is considered as a sus-
 224 tainable feedstock and can be used either energetically, chemically or materially (e.g. in
 225 the building sector). The latter two options are generally preferred over the energetic
 226 usage from a cleaner production perspective as biogenic carbon is stored in products

227 during their life span which contributes to climate change mitigation.

228 Despite substantial research on ecological advantages and pioneering initiatives to
229 establish AFS in practice, the share of AFS in agricultural production systems is still
230 small. E.g. in Germany about 1200 hectares of silvoarable AFS are established in 2025,
231 which accounts for about 0.02% of total agricultural land use [DEFAP]. One of the core
232 challenges in establishing AFS systems at large scale in practice is a lack of opportunities
233 to generate substantial revenues from woody biomasses of AFS. On the other side, the
234 rise of bioeconomy creates substantial demands for woody biomass and requires a reliable
235 and cost-efficient feedstock supply. As woody biomasses are bulky feedstocks with low
236 energy density, the design of cost-efficient supply chains is crucial to ensure the economic
237 viability of AFS-based value chains. This requires coordinated decisions on stand estab-
238 lishment, multi-cycle harvesting, and product cascading in spatially distributed supply
239 chains, which can be supported by integrated optimisation models that link stand-level
240 growth dynamics to regional supply chain design.

241 In the last two decades, a large body of research has analysed biomass supply chains
242 for bioenergy and biorefineries using mathematical programming models, often in the
243 form of mixed-integer linear programming (MILP) (Zahraee et al., 2020; Sharma et al.,
244 2013). These models typically optimise network configuration, feedstock sourcing, and
245 logistics under cost or profit objectives. They often treat biomass availability as an exoge-
246 nous input or rely on simplified agricultural production decisions. Moreover, downstream
247 processing of biomasses is concentrated on one particular value chain e.g. bioenergy plants
248 consuming all produced biomasses. Recent work has started to integrate operational de-
249 tails such as various biomass types, biomass growth models and regeneration processes
250 in multi-period supply chain models.

251 The present paper contributes to this literature in four ways. First, we propose
252 a novel MILP formulation for AFS-based biomass supply chain design that explicitly
253 links establishment decisions and consecutive harvest decisions, ensuring that harvesting
254 decisions at each site are consistent with minimum and maximum rotation ages. Second,
255 we embed allometric yield functions for different biomass types generating age-specific
256 yields for different woody biomass types (like stems, branches, or leaves and twigs). Third,
257 these different woody biomass types can be processed in specific downstream industries,

like chemical or paper industry as well as energy sector. The model aims to provide strategic decision support for establishing regional AFS-based bioeconomy ecosystems connecting multiple industries and product markets.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature on agroforestry biomass systems, allometric biomass modelling, and biomass supply chain optimisation. Section 3 summarises the MILP formulation for the agroforestry supply chain with age-lag constraints. Section ?? outlines the planned computational experiments for the case study and discusses expected insights for cleaner production strategies.

2. Literature review

Agroforestry biomass systems, stand-level growth models, and biomass supply chain optimisation have each been studied extensively, but they are rarely combined in an integrated framework that links age-dependent stand dynamics to regional product-cascading value chains (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). This section reviews (i) agroforestry systems with emphasis on system types, economics, and the cascading use of biomass as the primary revenue driver; (ii) biomass growth models at tree and stand level with explicit attention to the allometric–diameter–stand-biomass pathway; and (iii) biomass supply chain design using mathematical programming. The section concludes by identifying the modelling gap addressed in this paper.

2.1. Agroforestry systems

Agroforestry deliberately integrates woody perennials with crops and/or livestock on the same management unit, with the aim of enhancing productivity, profitability, and environmental performance (Nair, 1993; Toensmeier, 2017). In the temperate zone, three distinct system types can be distinguished based on the spatial arrangement of trees and the rotation length at which woody biomass is harvested.

Short-rotation coppice (SRC) is defined as a monoculture plantation of fast-growing woody species covering the entire management unit, with planting densities of 8,000–20,000 cuttings-ha⁻¹. After each harvest the root system regenerates without replanting. Rotation cycles are typically 3–7-years and total stand lifetime is 20–25-years

287 (Trnka et al., 2008; Faasch and Patenaude, 2012). SRCs are no AFSs in the narrow sense,
288 but it is helpful to benchmark biomass production.

289 **Short-rotation agroforestry systems (SRAFS)** integrate SRC-type tree strips
290 (typically 1–4 rows of the same fast-growing species like poplar or willow) into an otherwise
291 arable field in an alley-cropping arrangement. Crop production continues in the inter-
292 row strips (Huber et al., 2018; Graves et al., 2007). Thus, tree strips typically cover
293 10–30% of the total parcel area. Rotation lengths and total stand lifetime in SRAFS are
294 comparable to SRCs. In SRAFS all trees are effectively edge trees, exposed to higher light
295 and water availability than interior SRC trees. This boundary effect raises individual-tree
296 biomass accumulation but also intensifies competition with adjacent crop rows (Huber
297 et al., 2018).

298 **Long-rotation agroforestry systems (LRAFS)** use the same alley-cropping spa-
299 tial structure as SRAFS but with rotation cycles of 12–15~years or longer (Graves et al.,
300 2007; Nair, 1993). At this rotation length, individual trees develop substantial stem di-
301 ameters and the proportion of high-value timber increases markedly, enabling premium
302 material uses (veneer, sawlogs, engineered wood products) that are inaccessible in short-
303 rotation systems. Therefore, often timber-quality tree species (such as walnut, cherry,
304 oak, or high-grade poplar clones) are planted.

305 SRAFS and LRAFS are not clearly differentiated in practice. First, a clear cut-off age
306 is hard to define and, more importantly, the intended use of AFS biomass is not always
307 clear and can also change over time. Nonetheless, when establishing AFS, the structure
308 of AFS and its intended use after harvest drives planting decisions and, subsequently,
309 economic viability. Thus, in the present paper SRAFS and LRAFS are not considered as
310 separate AFS systems, but the interplay of intended use, establishment, and harvesting
311 decisions is analysed.

312 *2.1.1. Economic rationale of landowners*

313 The decision of a landowner to establish an AFS involves weighing substantial up-front
314 costs and long-term opportunity costs against a diversified revenue stream. On the **cost**
315 **side**, establishment requires site preparation, planting material, and planting labour,
316 typically amounting to 1,500–3,500 €/ha for SRC or SRAFS (Faasch and Patenaude,
317 2012; Testa et al., 2014). Recurrent harvest and refitting costs (machine hire, chipping,

transport to roadside) add approximately 200–400 €/ha per rotation depending on stand density and equipment availability (Testa et al., 2014; Spinelli et al., 2009). During the rotation, maintenance cost of AFS arise on a yearly basis. Next to expenses for establishment, maintenance and harvesting, opportunity cost arise as no revenues from crops are generated on tree strips. In regions with fertile soils and high cereal prices, yearly crop revenues may sum up to 400–600 €/ha (Faasch and Patenaude, 2012; Graves et al., 2007). Thus, opportunity cost are often the dominant cost driver in SRAFS profitability analyses.

On the **benefit side**, the primary revenue of AFS is the sale of harvested woody biomass, whose magnitude depends critically on rotation length, species, and site productivity. For poplar-based SRAFS in Central Europe, typical yields range from 10 to 25 t of dry matter biomass per ha for typical rotation lengths (Huber et al., 2018; Niemczyk and Thomas, 2021). Additionally, positive crop-yield effects in the inter-row strips of SRAFS and LRAFS may result. Tree rows act as windbreaks, reducing soil erosion and evapotranspiration; they provide partial shading that moderates heat stress during drought years; and their root systems may improve water retention in the topsoil (Graves et al., 2007; Grünewald et al., 2007). Empirical measurements in Central European alley-cropping trials indicate that these microclimate effects can increase cereal yields in adjacent strips by 5–15% relative to open-field controls, partially or fully compensating the loss of arable area occupied by the tree strips (Graves et al., 2007). Beyond direct yield effects, AFS deliver broader ecosystem services (like soil organic matter accumulation, carbon sequestration, biodiversity enhancement) that may lead to agri-environment subsidies and carbon credits (Toensmeier, 2017; Faasch and Patenaude, 2012).

Nonetheless, economic attractiveness of AFS depends primarily on revenues of AFS biomass. Thereby, revenues are different for the types of biomass and in which value chain it is used. Often AFS biomasses are used energetically, i.e. for heat and power generation, or anaerobic digestion, which typically creates low prices per tonne of biomass. Alternatively, woody parts of the biomass (i.e., the stem and large branches) can also be used as pulp and paper grade material as well as feedstock for chemical industry or as timber achieving higher prices (Lamers et al., 2011; De Meyer et al., 2014). Thus,

349 fostering AFS establishment by maximizing economic returns from AFS biomass requires
350 a strategic approach to product cascading that allocates different biomass fractions to
351 the highest-value applications, while ensuring that the rotation length is aligned with
352 the growth dynamics of the stand and the market demand for different biomass types
353 (Lamers et al., 2011).

354 Thus, for making educated decisions on AFS establishment and harvesting, the ex-
355 pected biomass quantities at harvest for stem, branches, and residues need to be esti-
356 mated. Thereby, the shares of the biomass types change with stand age. E.g.~the share
357 of stem weight rises from roughly 55% at age 4 to over 75% at age 10 (Huber et al., 2018;
358 Testa et al., 2014). Therefore, in Section 2.2 growth modelling approaches are reviewed
359 that capture the age-dependent growth dynamics of different biomass types in AFS.

360 2.2. Biomass growth models

Table 1: Notation for the biomass growth and harvest yield models.

Symbol	Description
$M^{sl}(t)$	Total above- and belowground dry biomass per ha of AFS (t DM ha ⁻¹) at stand age t
$A(N)$	Stand-level Gompertz asymptote (t DM ha ⁻¹); density- and site-dependent
k	Gompertz growth-rate coefficient (yr ⁻¹)
t_0	Inflection age of stand biomass accumulation (yr)
N	Stand density (trees ha ⁻¹)
C_{site}	Site-quality constant (kg ha ^{β-1}); conservative: 4475, good site: 6471
β	Competition-density exponent; fitted $\hat{\beta} = 0.380$
ϕ_{ag}	Aboveground fraction of total stand biomass; $\phi_{\text{ag}} = 0.89$
$M^{ag}(t)$	Aboveground dry biomass per ha (t DM ha ⁻¹) at age t
$f_p(t)$	asymtotic fraction of AGB attributed to compartment $p \in \{s, b, r\}$ at age t
α_p, δ_p	Intercept and slope of the linear fraction model for compartment p
$q_p(t)$	Merchantable fraction of compartment p biomass at age t (logistic)
r_p	Logistic growth rate for merchantability of compartment p
$t_{50,p}$	Age at which 50% of compartment p biomass is merchantable (yr)
$Y_p(t)$	Merchantable dry biomass of compartment p per ha at harvest age t (t DM ha ⁻¹)
ω	Moisture content at felling (mass fraction); $\omega = 0.55$

361 Reliable estimation of stand-level biomass quantities and their age-dependent compo-

sition is a necessary prerequisite for any strategic analysis of AFS-based supply chains. Individual-tree biomass is commonly approximated by species-specific allometric power laws relating shoot dry matter to stem base diameter, and these relationships have been calibrated for Central European poplar clones under SRC and SRAFS conditions (Huber et al., 2016, 2018). For supply chain planning, however, the relevant quantity is **stand-level biomass per hectare** over a multi-decade rotation horizon, which requires a temporal growth model that captures the characteristic sigmoidal accumulation trajectory of woody biomass.

2.2.1. Stand-level growth model

Stand-level aboveground and belowground dry biomass per hectare, $M^{sl}(t)$, accumulates in a right-skewed sigmoidal pattern with a relatively early inflection point, driven by rapid juvenile shoot growth followed by self-thinning and competitive deceleration as canopy closure intensifies (Niemczyk and Thomas, 2021; Testa et al., 2014). This trajectory is described by a Gompertz model:

$$M^{sl}(t) = A(N) \cdot \exp(-\exp(-k(t - t_0))), \quad (1)$$

where t_0 is the age of maximum instantaneous biomass increment and k determines how quickly the stand approaches its site-specific asymptote $A(N)$. For Central European poplar under medium-rotation AFS management (10–20 yr), calibration against empirical data from Huber et al. (2018), Niemczyk and Thomas (2021), and Jha (2018) yields $k = 0.194, \text{yr}^{-1}$ and $t_0 = 9.7, \text{yr}$ for a conservative site, and $k = 0.173, \text{yr}^{-1}$, $t_0 = 10.1, \text{yr}$ for a high-productivity site, placing the growth peak close to the biologically optimal rotation age identified by Niemczyk and Thomas (2021).

The asymptote $A(N)$ is not a fixed species constant but depends explicitly on planting density through a competition-density power law (?):

$$A(N) = \frac{C_{\text{site}} \cdot N^{1-\beta}}{1000}, \quad (2)$$

where C_{site} encodes site quality and β is the competition exponent. Equivalently, the per-tree asymptote $A_{\text{tree}}(N) = A(N) \cdot 1000/N = C_{\text{site}} \cdot N^{-\beta}$ decreases with density as individual trees have access to less growing space. Fitting (2) to two anchor points — a conservative AFS calibration at 113 trees, ha⁻¹ (central EU; $A = 84, \text{t, DM, ha}^{-1}$) and

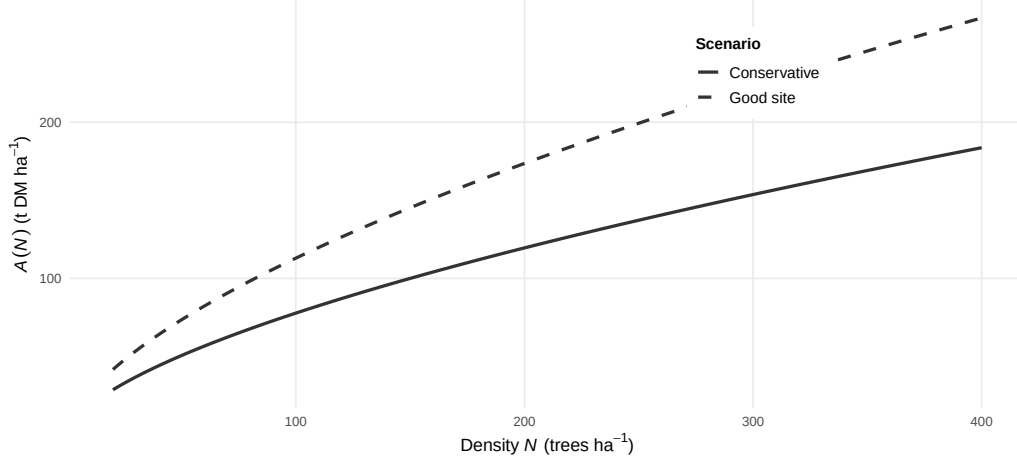


Figure 1: Stand-level Gompertz asymptote function $A(N)$ of planting density for the conservative (solid) and good-site (dashed) scenarios.

389 SRC data at 1,333 trees,ha⁻¹ from Niemczyk and Thomas (2021) — yields $\hat{\beta} = 0.380$,
 390 consistent with the 0.3–0.5 range reported for managed hardwood plantations (?). Two
 391 site-quality scenarios are used throughout: a **conservative scenario** ($C_{\text{site}} = 4\,475$, 113
 392 trees,ha⁻¹, $A = 84$,t,DM,ha⁻¹) representative of sandy loam soils in northern Central
 393 Europe, and a **good-site scenario** ($C_{\text{site}} = 6\,471$, 139 trees,ha⁻¹, $A = 138$,t,DM,ha⁻¹)
 394 representative of deep loamy soils in river valleys.

395 Within each compartment, only the biomass fraction whose element diameter exceeds
 396 a minimum threshold qualifies as merchantable product for industrial use. This fraction
 397 rises continuously with stand age as stems thicken and branches coarsen, and is modelled
 398 by a logistic function:

$$q_p(t) = \frac{f_p}{1 + \exp(-r_p(t - t_{50,p}))}, \quad (3)$$

399 where $t_{50,p}$ is the age at which 50% of compartment p biomass surpasses the diameter
 400 threshold. f_p denotes the asymptotic share of component p for mature trees. For the
 401 merchantable stem (threshold $d > 15$,cm), calibration to data from Civitarese et al. (2019)
 402 and Niemczyk and Thomas (2021) yields $r_{\text{stem}} = 0.467$,yr⁻¹, $t_{50,\text{stem}} = 8.19$,yr, and
 403 $f_{\text{stem}} = 0.75$. For merchantable branches (threshold $d > 7$,cm at insertion), Civitarese
 404 et al. (2019) and Jha (2018) imply $r_{\text{bran}} = 0.698$,yr⁻¹, $t_{50,\text{bran}} = 7.55$,yr, and $f_{\text{bran}} =$
 405 0.15.

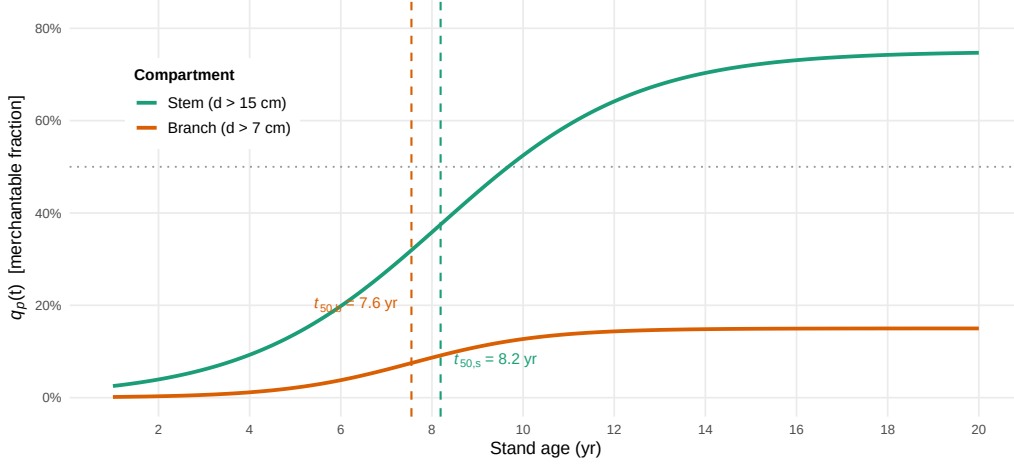


Figure 2: Logistic diameter-class merchantability functions $q_p(t)$ for stem ($d > 15$ cm, green) and branches ($d > 7$ cm, orange); the 50% threshold ages t_{50} are indicated by dashed vertical lines.

Combining equations (1)–(3), the merchantable dry biomass yield of compartment p per hectare at harvest age t is:

$$Y_p(t) = q_p(t) \cdot M^{ag}(t), \quad (4)$$

with all non-merchantable fractions aggregated into the residue stream. Fresh-weight yields at felling are obtained from dry matter yields via $Y_p^{fw} = Y_p / (1 - \omega)$, where the moisture content at felling $\omega = 0.55$ is taken from Civitarese et al. (2019). Figure 2 illustrates the shapes of $A(N)$, $f_p(t)$, and $q_p(t)$; Figure 3 shows the resulting stacked AGB component trajectories compared with empirical observations.

2.3. Biomass supply chain design

Biomass supply chain design problems have been extensively studied in literature (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). Most models jointly optimise facility locations and capacities, biomass sourcing patterns, multimodal transport flows, and inventory policies under cost or profit objectives, sometimes augmented by greenhouse gas emission indicators (Ekşioğlu et al., 2009; Arabi et al., 2019). A key methodological feature of the more advanced formulations is the integration of long-term network design with multi-period logistics, enabling simultaneous optimisation of infras-

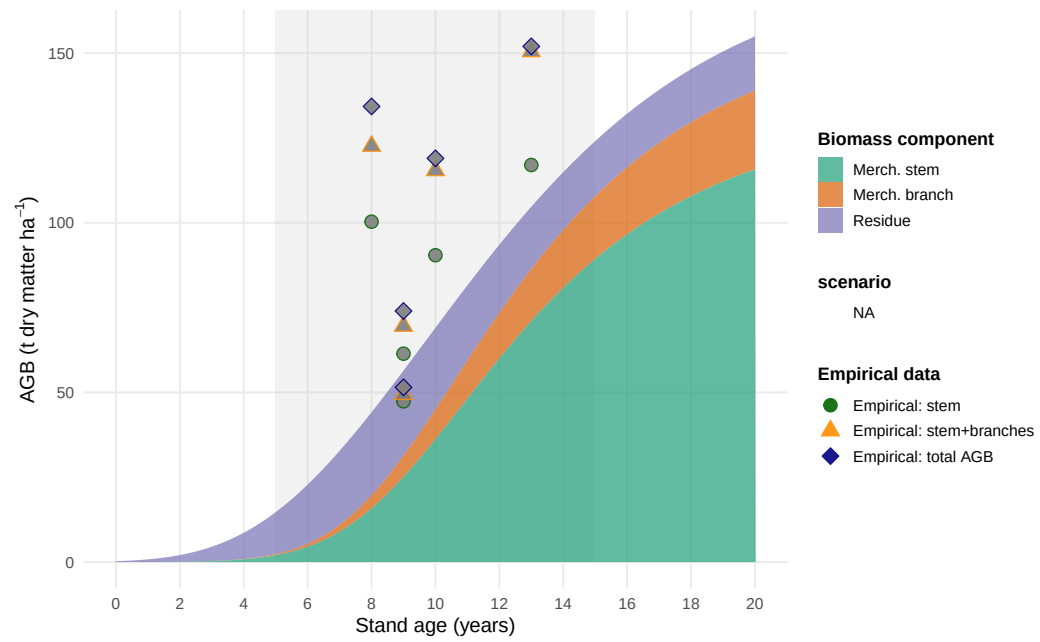


Figure 3: Stand-level aboveground biomass components for poplar under AFS management with $N = 250$ trees ha^{-1} . Empirical AGB observations compiled from Jha (2018), Civitarese et al. (2019), and Niemczyk and Thomas (2021). Grey band: typical medium-rotation harvest window (5–15 years).

421 tructure and operational decisions across seasonal or annual time steps (Ekşioğlu et al.,
422 2009; Zahraee et al., 2020).

423 Among generic multi-product frameworks, OPTIMASS explicitly models product
424 quality cascades by defining a hierarchy of biomass classes and allowing higher-quality
425 products to satisfy lower-quality demands, while the t-OPTIMASS extension incorpo-
426 rates biomass growth and regeneration over multiple periods (De Meyer et al., 2014,
427 2016). However, in virtually all existing formulations biomass availability is treated as
428 an exogenous time-series input: stand age structure, rotation decisions, and coppice cy-
429 cles are not represented endogenously, so that trade-offs between rotation length, stand
430 management, and supply chain configuration cannot be explored in a unified model
431 (De Meyer et al., 2016; Lamers et al., 2011). This gap is particularly pronounced for
432 agroforestry systems with repeated coppice cycles, where establishment timing, consec-
433 utive harvest decisions, and age-dependent biomass quality jointly determine both the
434 feasible supply profile and the achievable product cascade.

435 The MILP formulation developed in this paper addresses this gap by endogenising
436 stand age dynamics through binary age-lag variables, coupling the component-specific
437 yield coefficients η_{pa} derived from the Gompertz growth models above with supply chain
438 flow variables, and embedding a product quality cascade that allocates stem, branch,
439 and residue fractions to chemical, pulp, and energy markets within a single optimisation
440 model.

441 2.4. Biomass supply chain design

442 The design and planning of biomass supply chains has attracted substantial research
443 attention over the past two decades, driven by the need to ensure cost-efficient and
444 sustainable feedstock provision for bioenergy and biorefinery applications. Several sys-
445 tematic reviews provide comprehensive overviews of this literature. Cambero and Sowlati
446 (2014) survey economic, social, and environmental assessment and optimisation studies
447 for forest biomass supply chains, identifying MILP as the dominant modelling approach
448 and noting a growing trend towards multi-objective formulations integrating life cycle
449 assessment.

450 Yue et al. (2014) review biomass-to-bioenergy and biofuel supply chain models with
451 a focus on multi-scale modelling, uncertainty treatment, and sustainability metrics, and

conclude that the coupling of upstream feedstock dynamics with downstream network design remains an underexplored area. Malladi and Sowlati (2018) survey logistics-specific features of biomass supply chain models—seasonal availability, moisture content degradation, preprocessing steps—and highlight that most models treat biomass as a homogeneous commodity, neglecting quality differentiation. Zahraee et al. (2020) provide a recent meta-review of 140+ studies, confirming that cost minimisation is still the dominant objective, and that multi-product cascading formulations are rare.

The structure of biomass supply chain models typically encompasses three decision layers: (i) strategic network design (facility location, capacity sizing, technology selection), (ii) tactical planning (harvest scheduling, seasonal flows, inventory), and (iii) operational routing. The following paragraphs discuss key individual contributions and their specific methodological advances.

Ekşioğlu et al. (2009) develop one of the earliest large-scale MILP models for biomass-to-biorefinery supply chain design, integrating facility location, capacity sizing, and multimodal transport for lignocellulosic feedstocks. Their model minimises total system cost and introduces binary facility-opening variables that have since become standard in the field. A key limitation, however, is that biomass supply is treated as an exogenous input, so feedstock availability is not endogenised as a function of land-use or growth decisions. Mansoornejad et al. (2013) extend the strategic design perspective to forest biorefineries by integrating product portfolio design with supply chain configuration in a single MILP. Their model explicitly accounts for product conversion efficiencies along different processing pathways, enabling joint optimisation of process technology selection and logistics.

De Meyer et al. (2014) introduce the OPTIMASS framework, a generic MILP for biomass-based supply chain optimisation that explicitly models product quality cascades: higher-quality biomass classes can substitute for lower-quality demands downstream, whereas the reverse is infeasible. This cascade hierarchy, formalised through ordered product sets, is particularly relevant for woody biomass where stem, branch, and residue fractions command different market prices. The companion t-OPTIMASS extension (De Meyer et al., 2016) incorporates temporal dynamics by explicitly representing biomass growth and regeneration over multiple periods, enabling simultaneous

optimisation of harvest timing and network flows. However, in both OPTIMASS formulations the growth dynamics are represented as exogenous time-series rather than as age-structured stand models, so rotation-length decisions and age-dependent biomass quality are not endogenised. Arabi et al. (2019) demonstrate multi-period MILP extensions for bio-based supply chains with production dynamics, but focus on microalgae rather than woody biomass and do not address product quality cascades.

Sharma et al. (2013) identify supply-side uncertainty (yield variability, seasonal availability, moisture content) as the most critical source of risk in biomass supply chains. In response, a strand of literature has developed stochastic and robust optimisation extensions. Zahraee et al. (2020) document that two-stage stochastic MILP and robust optimisation approaches now account for roughly 30% of the modelling literature, with biomass yield uncertainty and price volatility as the dominant stochastic parameters.

Biomass supply chain research focusing specifically on agroforestry or SRC contexts is comparatively sparse. Espinoza et al. (2017) note that most supply chain models assume continuous feedstock availability and do not represent the periodic harvest cycles characteristic of coppice systems. Lamers et al. (2011) analyse feedstock design principles for lignocellulosic supply chains and argue that the allocation of different biomass fractions to valorisation pathways—product cascading—is a key lever for economic viability, yet is rarely optimised jointly with network design decisions. The model developed in Section~3 of this paper directly addresses this gap by endogenising stand age dynamics through binary age-lag variables and coupling component-specific yield coefficients with a product quality cascade.

Table~2 summarises the key modelling features of the reviewed contributions.

3. MILP Model for Agroforestry Supply Chain Design

Building on the biomass growth models introduced in Section 2.2 and the supply chain design literature reviewed in Section 2.3, this section describes the AFS supply chain design problem (AFS-SCD), a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain. The model endogenises stand age dynamics through binary age-lag variables, couples the component-specific yield coefficients η_{pa} derived from the Gompertz growth models (??)–

Table 2: Key modelling features of selected biomass supply chain design studies.

Study	Objective	Multi-product	Quality cascade	Age dynamics	Uncertainty	Level
Eksioglu et al. (2009)	Cost min.	—	—	—	—	Strategic
Sharma & Taaffe (2013)	Cost min.	Partial	—	—	—	Strategic/Tactical
Mansoornejad et al. (2013)	Profit max.	✓	—	—	—	Strategic
Cambero & Sowlati (2014)	Cost/GHG/jobs	—	—	—	Stochastic	Strategic
Yue et al. (2014)	Cost/GHG	✓	—	—	—	Multi-scale
De Meyer et al. (2015)	Cost min.	✓	✓	Exogenous	—	Strategic/Tactical
De Meyer et al. (2016)	Cost min.	✓	✓	Exogenous TS	—	Multi-period
Lamers et al. (2015)	Cost/quality	✓	Partial	—	Scenario	Strategic
Espinoza et al. (2017)	Sustainability	✓	—	—	Scenario	Strategic
Malladi & Sowlati (2018)	Cost min.	Partial	—	—	—	Tactical/Oper.
Arabi et al. (2019)	Profit max.	✓	—	Exogenous	—	Multi-period
Zahraee et al. (2020)	Multi-obj.	✓	—	—	Stochastic	Strategic/Tactical
This paper	NPV max.	✓	✓	Endogenous	Scenario	Strategic/Multi-period

(??) with supply chain flow variables, and assigns AFS biomass types (i.e., stem, branch, and residue fractions) to potential consumers in chemical, pulp and paper as well as energy industry.

3.1. Sets, Indices, and Time Structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/pre-treatment facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and biomass types $\mathcal{P} = \{\text{stem}, \text{branches}, \text{residues}\}$, whose growth models are outlined above in (??)–(??). Time is discretised into annual periods $t \in \mathcal{T} = \{1, \dots, T_{\max}\}$. To model flexible harvest timing according to typical rotation structures of SRAFS and LRAFS, $A^{\min}$ and $A^{\max}$ define minimum and maximum stand ages. Thus, harvest are possible in periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$. To consistently model establishment and consecutive harvest decisions, the following arc sets are defined: $\mathcal{S}^{\text{est}}$ defines potential establishments of AFS. If an AFS is established in period $t \in \mathcal{T}$, arc $(0, t)$ is selected. Harvest set $\mathcal{S}^{\text{harv}}$ summarizes all potential consecutive harvest periods

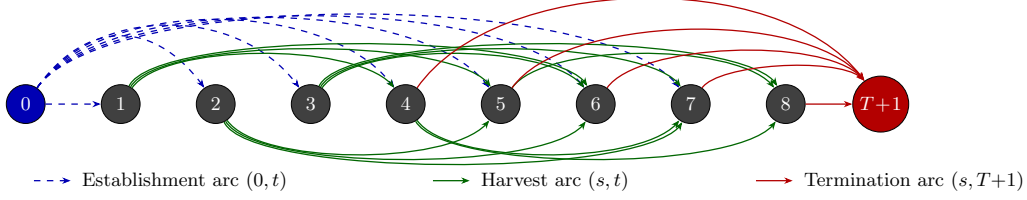


Figure 4: Network structure of set $\mathcal{S}$ for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$. Dashed blue arcs: establishment arcs $(0, t)$, $t \in \{1, \dots, T-1\}$. Solid green arcs above: harvest arcs (s, t) with odd s ; below: harvest arcs with even s . Solid red arcs: termination arcs $(s, T+1)$, $s \in \{A_{\min} + 1, \dots, T\}$.

within the age-lag bounds $A^{\min}$ and $A^{\max}$. I.e., if an AFS is established or harvested in period s and next harvest is in period t , arc (s, t) is selected. Finally, there is no subsequent harvest after period t , the AFS is terminated by selecting arc $(t, T^{\max} + 1)$. Termination arcs are summarized in set $\mathcal{S}^{term}$. All arc sets are formally defined as follows:

$$\begin{aligned}\mathcal{S}^{est} &= \{(0, t) \mid t \in \{1, \dots, T^{\max} - A^{\min}\}\}, \\ \mathcal{S}^{harv} &= \{(s, t) \mid s, t \in \mathcal{T} \wedge t - s \in \{A^{\min}, \dots, A^{\max}\}\}, \\ \mathcal{S}^{term} &= \{(s, T^{\max} + 1) \mid s \in \{A^{\min} + 1, \dots, T_{\max}\}\}, \\ \mathcal{S} &= \mathcal{S}^{est} \cup \mathcal{S}^{harv} \cup \mathcal{S}^{term}.\end{aligned}$$

Figure 4 illustrates this network structure of all arc sets for $T = 8$, $A^{\min} = 3$, $A^{\max} = 5$.

3.2. Decision Variables

Binary variables z_{ist} indicate whether an arc from set $(s, t) \in \mathcal{S}$ is selected for site i , indicating establishment, (consecutive) harvest, or termination. Taken together across all arcs, these variables trace a path of consecutive harvest cycles at each site over the planning horizon. Figure 5 illustrates a feasible path corresponding to the establishment of an AFS in period 1 and harvests in periods 5 and 8, whereby the harvest in $t = 8$ is final (and AFS is terminated, thus).

When a AFS is harvested in period t , continuous variables $Y_{ipt} \geq 0$ represent the harvest quantity (t) of product p at site i . Flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from AFS sites to storage/pre-treatment facilities and from there to consumers, respectively, while S_{jpt} captures end-of-period inventory at facility j .

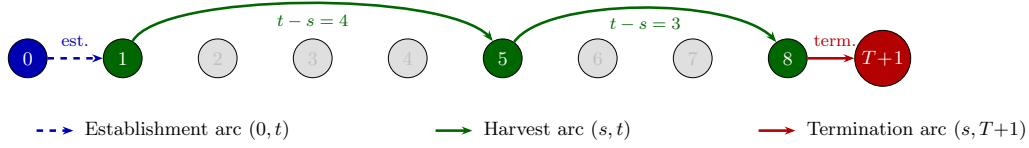


Figure 5: Example of a feasible path through the network for $T = 8, A_{\min} = 3, A_{\max} = 5$: establishment in period 1, harvests in periods 5 and 8, termination at $T + 1$.

Table 3 summarises all sets, decision variables, and parameters used in the MILP formulation. Symbols shared with the growth model of Section 2.2 — in particular the yield coefficients η_{pa} , which are derived from the component-specific Gompertz functions (??)–(??) and the stand biomass model (??) — are listed in Table ?? in Section 2.2.

3.3. Objective Function

As discussed in Section 2.1, the economic attractiveness of AFS supply chains must be assessed from a global perspective taking into account all processes, activities, and participants involved in finally serving customer markets with bio-based feedstocks. Therefore, revenues for the different biomass fractions and must be weighed against costs occurring at all value-adding stages of the supply chains. To be precise, revenues created by selling AFS-based feedstocks to different industrial consumers are counterweighted by establishment, opportunity, harvest, transport, and storage costs of landowners, agricultural firms as well as biomass logistics companies. The model therefore maximises total contribution

margin of all participants in the AFS supply chain over the planning horizon as follows:

$$\begin{aligned}
\max Z = & \sum_{k \in \mathcal{K}, p \in \mathcal{P}} R_{kp} \cdot \sum_{j \in \mathcal{J}, (p', p) \in \mathcal{Q}, t \in \mathcal{T}} X_{jkp't} \\
& - \sum_{i \in \mathcal{I}} c_i^{est} \cdot AREA_i \cdot \sum_{(0, t) \in \mathcal{S}^{est}} z_{i0t} \\
& - \sum_{i \in \mathcal{I}} c_i^{opp} \cdot AREA_i \cdot \left(\sum_{(s, T+1) \in \mathcal{S}^{term}} (s+1) \cdot z_{is(T+1)} - \sum_{(0, t) \in \mathcal{S}^{est}} t \cdot z_{i0t} \right) \\
& - \sum_{i \in \mathcal{I}} c_i^{main} \cdot (t-s) \cdot AREA_i \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
& - \sum_{i \in \mathcal{I}} c_i^{harv} \cdot AREA_i \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
& - \sum_{i \in \mathcal{I}, j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}} c_p^{tr-raw} \cdot d_{ij} \cdot X_{ijpt} \\
& - \sum_{j \in \mathcal{J}, k \in \mathcal{K}, (p, p') \in \mathcal{Q}, t \in \mathcal{T}} c_{p'}^{tr-pre} \cdot d_{jk} \cdot X_{jkpp't} \\
& - \sum_{j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}} c_j^{stor} \cdot S_{jpt}
\end{aligned} \tag{5}$$

Here, R_{kp} are product- and consumer-specific revenues (€/t). The opportunity cost term penalises dereduced crop revenues in each year an AFS is established. Transport cost terms mirror the two-stage flow structure of the AFS supply chain with differentiated rates for raw and pre-treated biomass depending on the biomass type. Cost rates differ because as e.g. the density and water content of the shipped biomasses differ or because specialized equipment (e.g. for log transports) is used [REFERENCES]. Moreover, site-specific establishment cost for planting the AFS trees is considered, next to yearly maintenance costs and harvesting cost. All these costs scale with the size of a site ($AREA_i$) measured in hectar. Maintenance cost comprise e.g. costs for trimming, replanting, and pruning unnecessary shoots. Storage costs are considered if biomasses are stored for more than a year to compensate direct costs (e.g. for storage site maintenance) but primarily opportunity cost due to potential material losses and capital commitment. The usual case is that AFS are harvested at the beginning of a year. Harvested biomass dry during spring and summer such that the biomasses can be used in autumn or winter the same year. As drying is necessary for most industrial applications and beneficial for transport economics, drying storage in the year of harvest is not associated with

573 additional cost.

574 3.4. Constraints

575 Constraints (6) limit each site to at most one establishment decision over the planning
576 horizon. Constraints (7) enforce flow conservation in each period. If an activity in period
577 $t \in \mathcal{T}$ takes place, there needs to be a preceeding activity (harvest or establishment) and
578 a succeeding activity (harvest or termination). This way, (7) guarantee valid (possibly
579 empty) activity paths over the complete time horizon for each site. Constraints (8) links
580 the binary harvest-cycle variables to physical yield quantities via the age-dependent yield
581 coefficients $\eta_{p(t-s)}$. These coefficients are pre-computed by evaluating the component-
582 specific Gompertz functions (??) and (??) at stand age $a = t - s$ and scaled by site
583 area AREA_i . Thus, implicitly we assume constant yields per hectare for each site i . Note
584 that this simplification can be dropped and site-specific yields $\eta_{ip(t-s)}$ could be intro-
585 duced without increasing numerical complexity of the AFS-SCD . Constraints (??)–(9)
586 link harvest quantities to transport flows and ensure inventory balance at each facility,
587 following the multi-period network-flow structure of OPTIMASS (De Meyer et al., 2014,
588 2016). Thereby, harvest quantities can be smaller than available AFS biomasses allowing
589 partial harvesting. Constraints (10)–(11) enforce storage and processing capacity limits
590 at each facility j . Constraints (12) implement product usage at multiple consumption
591 sites k . Thereby, $(p', p) \in \mathcal{Q}$ defines a product mapping allowing high-quality product
592 p' (e.g. stems) satisfying demand of a lower-quality products p up to maximum demand
593 $D_{kpt}^{\max}$. This way, the cascading principle is operationalized as a central economic ratio-
594 nale.

$$\sum_{(0,t) \in \mathcal{S}^{est}} z_{i0t} \leq 1 \quad \forall i \in \mathcal{I} \quad (6)$$

$$\sum_{(s,t) \in \mathcal{S}} z_{ist} = \sum_{(t,u) \in \mathcal{S}} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (7)$$

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq \sum_{(s,t) \in \mathcal{S}^{harv}} \eta_{p(t-s)} \cdot \text{AREA}_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (8)$$

$$S_{jpt} = S_{jp,t-1} + \sum_{i \in \mathcal{I}} X_{ijpt} - \sum_{k \in \mathcal{K}, (p,p') \in \mathcal{Q}} X_{jkpp't} \quad \forall j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (9)$$

$$\sum_{p \in \mathcal{P}} S_{jpt} \leq \text{CAP}_j^{stor} \quad \forall j \in \mathcal{J}, t \in \mathcal{T} \quad (10)$$

$$\sum_{i \in \mathcal{I}, p \in \mathcal{P}} X_{ijpt} \leq \text{CAP}_j^{proc} \quad \forall j \in \mathcal{J}, t \in \mathcal{T}^{harv} \quad (11)$$

$$\sum_{j \in \mathcal{J}, (p',p) \in \mathcal{Q}} X_{jkp'pt} \leq D_{kpt}^{\max} \quad \forall k \in \mathcal{K}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (12)$$

$$z_{ist} \in \{0, 1\} \quad \forall i \in \mathcal{I}, (s, t) \in \mathcal{S} \quad (13)$$

$$Y_{ipt}, X_{ijpt}, X_{jkp'pt}, S_{jpt} \geq 0 \quad \forall i, j, k, p, p', t \quad (14)$$

595 The yield coefficients η_{pa} in (8) are the critical bridge between the stand-level growth
 596 model of Chapter~2 and the supply chain optimisation. For stand age $a = t - s$ and
 597 product component p , they are computed as:

$$\eta_{pa} = f_p(a) \cdot M^{sl}(a) = \frac{h_p(a)}{\sum_{p'} h_{p'}(a)} \cdot a^\infty \cdot e^{-e^{-b'^1 \cdot (a-\tau)}}, \quad (15)$$

598 where $M^{sl}(a)$ follows from (??) and $f_p(a)$ follows from (??). For poplar clone Max~3 un-
 599 der SRAFS conditions, the component-specific Gompertz parameterisations from Section
 600 2.2 imply that $\eta_{stem,a}$ rises steeply between ages~4 and 10, while $\eta_{branch,a}$ and $\eta_{residue,a}$
 601 plateau earlier, as illustrated in Figure~??. This age-dependence of product fractions is
 602 the core mechanism through which rotation-length decisions — governed by $A^{\min}$ and
 603 $A^{\max}$ and the arc set $\mathcal{S}$ — determine achievable product portfolios and, hence, revenues
 604 in the objective function (5).

605 4. Case Study

606 The proposed MILP model is intended to support the design of agroforestry biomass
607 supply chains for cleaner production in specific regional contexts. In a forthcoming case
608 study, we focus on poplar-based alley-cropping systems in a temperate European setting,
609 informed by empirical data from SRAFS trials in Southern Germany and SRC plantations
610 in Central Europe (Huber et al., 2018; Trnka et al., 2008; Testa et al., 2014).

611 4.1. Study Region and Supply Chain Configuration

612 The proposed AFS-SCD model is instantiated on a real-world biomass supply chain lo-
613 cated in the tri-state region of Saxony-Anhalt, northern Thuringia, and northern Saxony,
614 Germany (Figure 6). This region combines the highest density of registered short-rotation
615 coppice (SRC) Feldblöcke in eastern Germany with a cluster of major wood-processing
616 industries that collectively represent one of the largest forest-based biomass demand cen-
617 tres in Central Europe. The spatial extent of the study region covers approximately
618 250×200 km, with road distances between potential KUP sites and consumer locations
619 ranging from under 10 km to over 180 km.

620 The supply chain comprises three node types: (i) KUP production sites $\mathcal{I}$, (ii) pre-
621 treatment and intermediate storage facilities $\mathcal{J}$, and (iii) consumer sites $\mathcal{K}$. Candidate
622 KUP sites are drawn from the official Feldblock register of Saxony-Anhalt, filtered to
623 parcels with a minimum area of 1 ha and classified as potentially suitable for SRC
624 under the InVeKoS land-use scheme. Following the two-stage spatial clustering procedure
625 described in Section-??, the $|\mathcal{I}| = 2239$ raw Feldblöcke are grouped into K super-sites
626 using DBSCAN ($\varepsilon = 15$ km, $minPts = 3$) and Ward.D2 hierarchical clustering (maximum
627 within-cluster radius 15 km), yielding a computationally tractable MILP instance while
628 preserving the total eligible area. The HAC cluster assignments, visible as distinct colour
629 zones in Figure 6, confirm a spatially coherent partitioning: dense KUP corridors along
630 the Elbe and Saale floodplains form contiguous super-sites, while isolated parcels in the
631 Harz foothills constitute smaller or singleton clusters.

632 Two types of intermediate infrastructure nodes are considered: conventional sawmills
633 and logistics hubs. Sawmills perform primary mechanical processing (debarking, chip-
634 ping) and can serve product streams P1 and P2; logistics hubs serve as consolidation

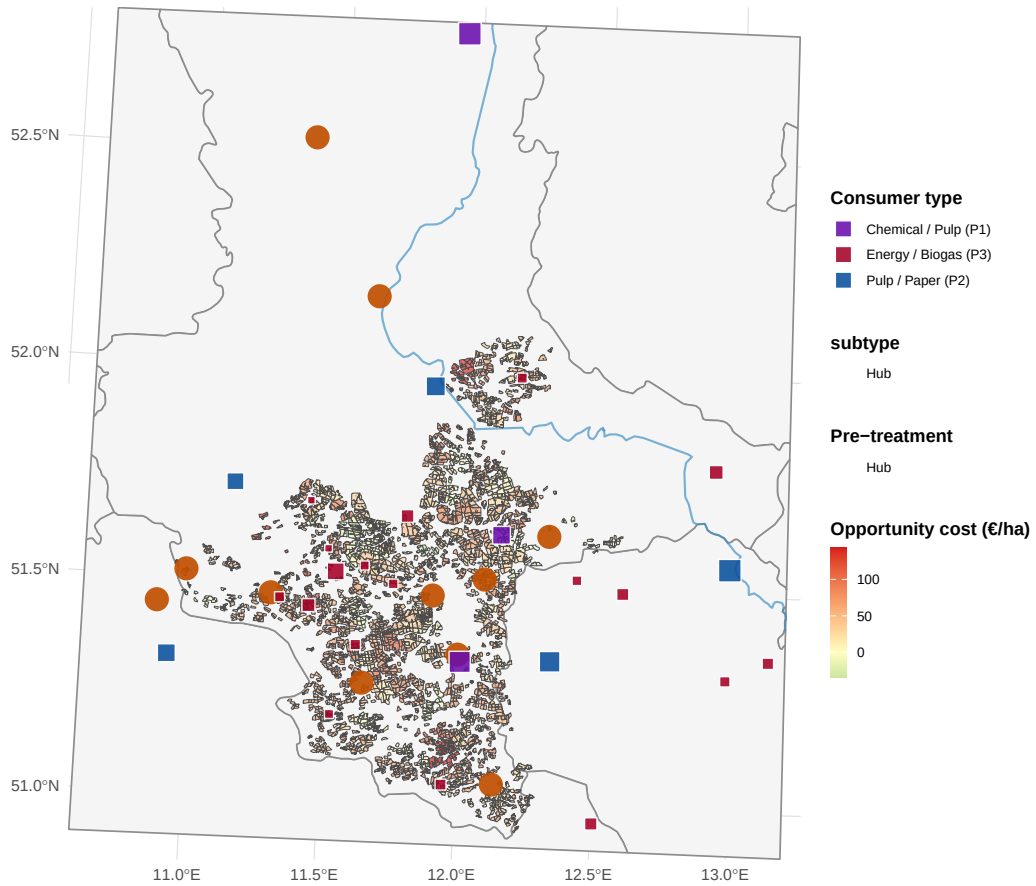


Figure 6: AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Coloured polygons: potential KUP/AFS sites grouped by HAC super-site (colours encode cluster membership; no legend). Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

and temporary storage points for all three product grades and are characterised by lower capital intensity but limited processing capacity. The $|\mathcal{J}| = 11$ candidate storage/pre-treatment locations are sourced from publicly registered forestry enterprises and existing logistic nodes in the study region. Road distances d_{ij} (site i to storage j) and d_{jk} (storage j to consumer k) are computed via the OSRM open-source routing engine using the German road network, providing realistic transport cost estimates that are embedded directly in the MILP objective.

4.2. Consumer Demand and Product Classification

Nine consumer sites are included in the base case ($|\mathcal{K}| = 9$), covering the full quality cascade from high-value chemical/pulp applications down to energy use (Table 4). Consumer demands $D_{kpt}^{\max}$ are derived from publicly reported annual throughput figures, production capacities, and regulatory filings, and are held constant across periods ($D_{kpt}^{\max} = D_{kp}^{\max} \forall t \in \mathcal{T}$) to reflect stable long-term offtake contracts.

Three biomass product grades are distinguished, reflecting the cascading-use principle embedded in the AFS-SCD objective:

- **P1 – Chemical/pulp grade** ($p = 1$): long-fibre hardwood chips suitable for chemical pulping and biorefinery conversion. Dominated by Mercer Stendal GmbH (Arneburg, 1{,}500 kt/yr) and Zellstoff Rosenthal/Mercer (Blankenstein, 1{,}000 kt/yr), which together account for 403% of total P1 demand. UPM Biochemicals (Leuna) contributes 500 kt/yr of high-purity beech-fraction demand for biochemical conversion. Gate prices for P1 range from 80 to 120 €/t wood input.
- **P2 – Sawlog/pulpwood grade** ($p = 2$): roundwood and chipped material suitable for mechanical processing in sawmills. Demand is distributed across six smaller facilities with individual annual demands of 5–300 kt/yr. Gate prices range from 55 to 75 €/t.
- **P3 – Energy/pellets grade** ($p = 3$): residual wood fractions (chips, bark, off-cuts) used for heat and electricity generation or pellet production. This grade absorbs material that cannot meet P1 or P2 quality thresholds, operationalising the cascade constraint in the MILP. Gate prices range from 30 to 40 €/t.

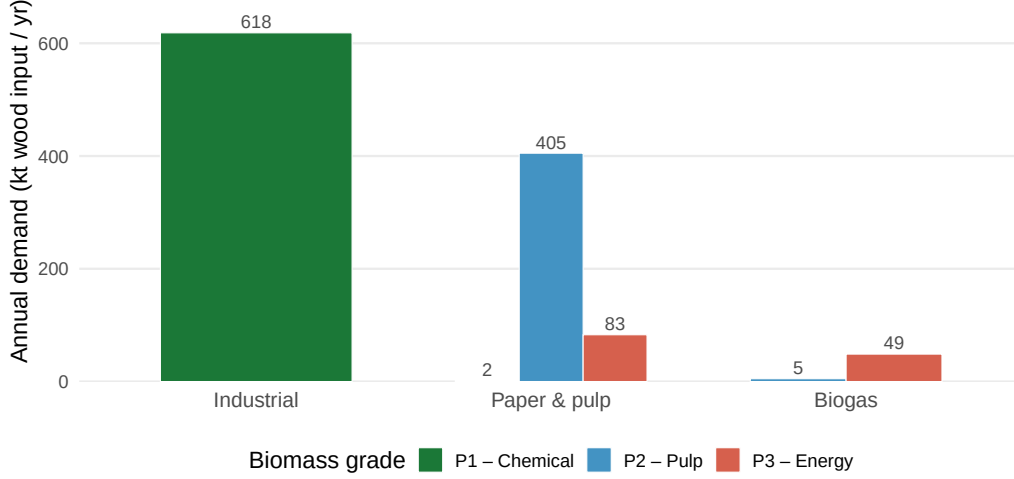


Figure 7: Total biomass demand by consumer type and grade (P1 chemical, P2 pulp & paper, P3 energy/biogas). Bars show aggregated annual demand in kt wood input per year across all individual plants within each consumer type.

664 The demand derivation follows a three-step procedure. First, publicly available ca-
 665 pacity data (kt wood input per year) are collected for each facility from official sources,
 666 industry reports, and corporate disclosures (Mercer Stendal GmbH, 2024; UPM Bio-
 667 chemicals, 2024). Second, the total annual capacity is assigned to the product grade that
 668 corresponds to the primary feedstock of each facility (P1 for chemical/biorefinery pro-
 669 cesses, P2 for mechanical sawmilling, P3 for thermal conversion and pellet lines). Third, a
 670 minor P3 component is added to sawmill facilities to capture by-product streams (bark,
 671 sawdust), consistent with observed industrial practice. The resulting demand matrix
 672 $D_{kp}^{\max}$ (kt/yr) is shown in Figure 7 and summarised in Table 4. Aggregate demand across
 673 the study region totals 620.5 kt/yr for P1, 410 kt/yr for P2, and 131.7 kt/yr for P3.

674 4.3. MILP Model Parameters

675 Table 5 summarises all scalar parameters of the AFS-SCD instance. The planning
 676 horizon covers $|\mathcal{T}| = 15$ periods, corresponding to one full poplar rotation (years 1–15),
 677 with a minimum harvest age $a^{\min} = 4$ and maximum productive stand age $a^{\max} =$
 678 15 (Lehmkuhl et al., 2023). Biomass yield functions η_{pa} (odt ha⁻¹ yr⁻¹) follow the
 679 age-structured Chapman-Richards profile calibrated in Section~??, where distinct yield

680 curves are assigned to each product grade to reflect the fraction of stem, branch, and
681 foliage biomass that qualifies under each quality threshold.

682 Transport costs are disaggregated into two components reflecting the two-echelon
683 logistics structure. Raw material transport from site i to pre-treatment facility j incurs
684 a variable cost $c^{tr-raw} \cdot d_{ij}$ ($\text{€ t}^{-1} \text{ km}^{-1}$), while processed material transport from j to
685 consumer k is costed at $c^{tr-pre} \cdot d_{jk}$. The pre-treatment rate c^{tr-pre} is set lower than c^{tr-raw}
686 to reflect the higher load density and standardised packaging of chipped and pelletised
687 material compared to loose roundwood. An opportunity cost parameter c^{opp} (€ ha^{-1}
688 yr^{-1}) accounts for foregone agricultural revenue on converted parcels, calibrated to the
689 average arable land rental rate in Saxony-Anhalt (Statistisches Bundesamt (Destatis),
690 2023). Storage capacity at each facility j is bounded by $\bar{S}_j$, and processing capacity by
691 $\bar{C}_j$ (kt yr^{-1}), both drawn from publicly available facility data.

692 4.4. Optimization results

693 5. Conclusion

694 6. Appendix

695 References

- 696 Mohsen Arabi, Saeed Yaghoubi, and Hamed Tajik. Optimal design of a microalgae-based biofuel supply
697 chain with co2 reduction and hydrogen production. *Computers & Chemical Engineering*, 130:106530,
698 2019. ISSN 0098-1354. doi: 10.1016/j.compchemeng.2019.106530.
- 699 Claudia Cambero and Taraneh Sowlati. Assessment and optimization of forest biomass supply chains
700 from economic, social and environmental perspectives—a review of literature. *Renewable and Sus-*
701 *tainable Energy Reviews*, 36:62–73, 2014. ISSN 1364-0321. doi: 10.1016/j.rser.2014.04.041.
- 702 Vincenzo Civitarese, Andrea Acampora, Giulio Sperandio, Alberto Assirelli, and Rodolfo Picchio. Pro-
703 duction of wood pellets from poplar trees managed as coppices with different harvesting cycles. *En-*
704 *ergies*, 12(15):2973, 2019. doi: 10.3390/en12152973. Open access.
- 705 Anouck De Meyer, Dirk Cattrysse, Jussi Rasinmäki, and Jos Van Orshoven. Methods to optimise the
706 design and management of biomass-for-bioenergy supply chains: A review. *Renewable and Sustainable*
707 *Energy Reviews*, 31:657–670, 2014. ISSN 1364-0321. doi: 10.1016/j.rser.2013.12.036.
- 708 Anouck De Meyer, Dirk Cattrysse, and Jos Van Orshoven. A generic mathematical model to optimise
709 strategic and tactical decisions in biomass-based supply chains (OPTIMASS). *European Journal of*
710 *Operational Research*, 249(1):293–307, 2016. ISSN 0377-2217. doi: 10.1016/j.ejor.2015.08.003.

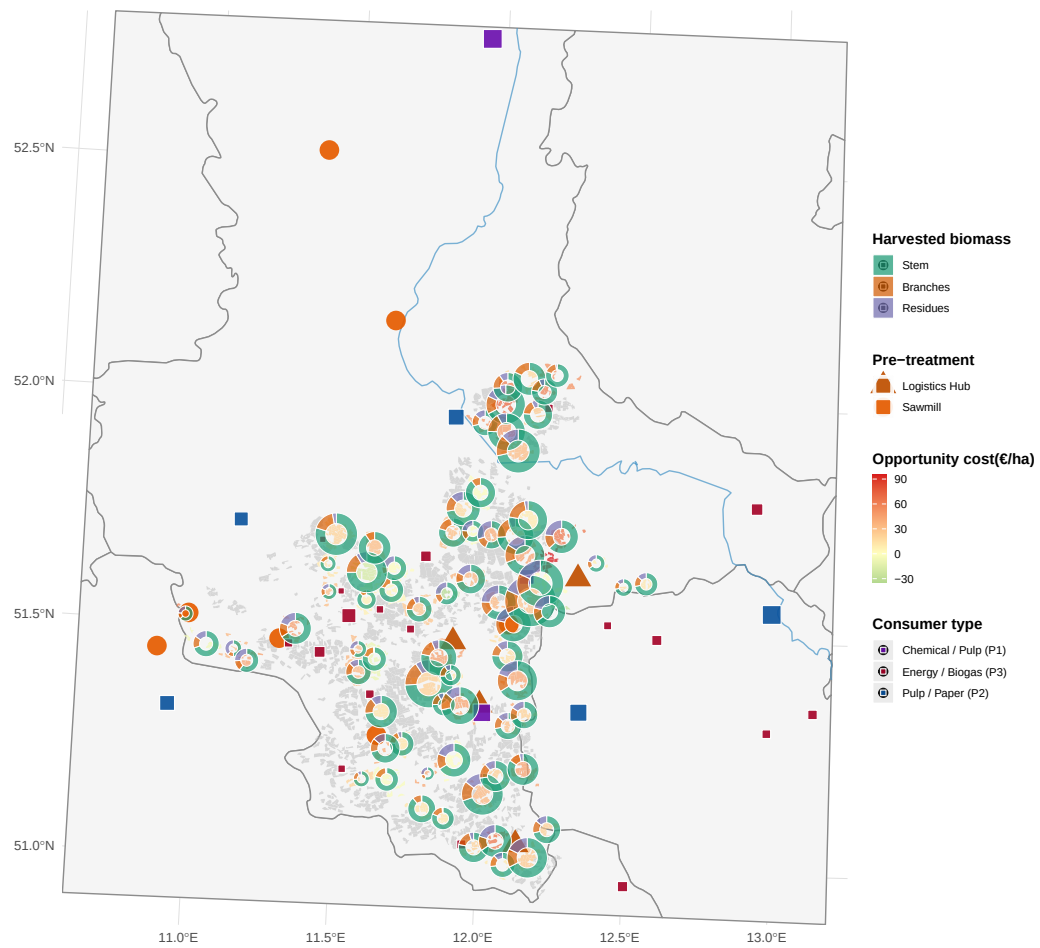


Figure 8: Optimal agroforestry supply chain network. Green polygons: active super-sites with biomass harvest; grey polygons: non-selected sites. Pie charts at active sites show the product mix of harvested biomass (stem = P1, branches = P2, residues = P3), scaled by total harvested volume. Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub). Coloured squares: consumer sites by dominant demand category.

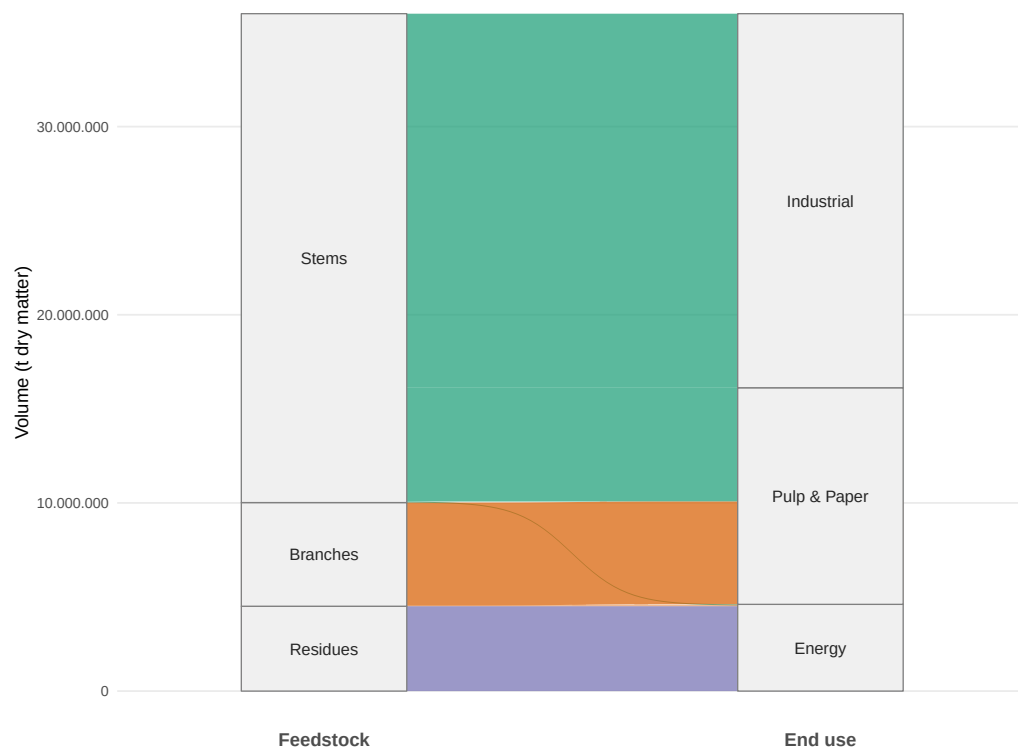


Figure 9: Total production quantity of biomass types and its industrial usage over planning horizon (in tons).

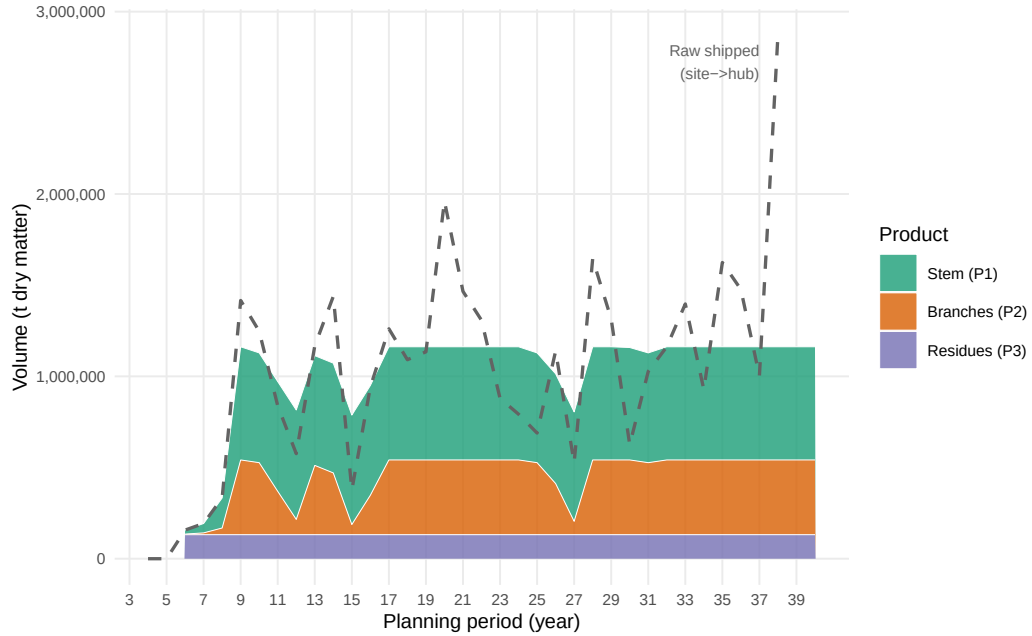


Figure 10: Total biomass delivered to consumers by product type over the planning horizon. Stacked areas show period-level volumes of stem (P1), branches (P2), and residues (P3) flowing from storage hubs to all consumers. Dashed line: total harvested raw biomass shipped from sites to hubs.

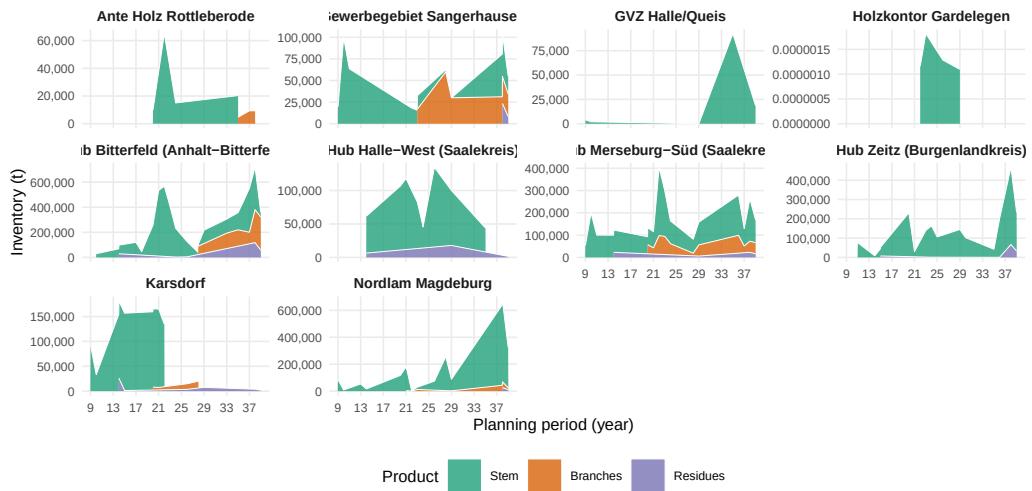


Figure 11: End-of-period inventory levels at storage/pre-treatment hubs, disaggregated by product type. Each panel corresponds to one active hub.

- 711 Sandra Duni Ekşioğlu, Aditya Acharya, Lawrence E. Leightley, and Sumit Arora. Analyzing the design
712 and management of biomass-to-biorefinery supply chain. *Computers & Industrial Engineering*, 57(4):
713 1342–1352, 2009. ISSN 0360-8352. doi: 10.1016/j.cie.2009.07.003.
- 714 Victor Espinoza, Selene Bautista, Patricia C. Narvaez, Mauricio Alfaro, and Mauricio Camargo. Sustain-
715 ability assessment to support governmental biodiesel policy in Colombia: A system dynamics model.
716 *Journal of Cleaner Production*, 141:1145–1163, 2017. ISSN 0959-6526. doi: 10.1016/j.jclepro.2016.09.
717 168.
- 718 Ralf J. Faasch and Gaia Patenaude. The economics of short rotation coppice in Germany. *Biomass and*
719 *Bioenergy*, 45:27–40, 2012. ISSN 0961-9534. doi: 10.1016/j.biombioe.2012.04.012.
- 720 Anil R. Graves, Paul J. Burgess, Fabrice Liagre, Andrea Pisanelli, Patrice Paris, Gerardo Moreno,
721 Manuel Bellido, Manuela Mayus, Meindert Postma, Bernhard Schindler, Konstantinos Mantzanas,
722 and Christian Dupraz. Development and application of bio-economic modelling to compare silvoarable,
723 arable, and forestry systems in three European countries. *Ecological Engineering*, 29(4):434–449, 2007.
724 ISSN 0925-8574. doi: 10.1016/j.ecoleng.2006.09.018.
- 725 Henning Grunewald, Reinhold Böcker, B. Ulrich Schneider, Oliver Bens, and Reinhard F. Hüttl. Biomass
726 production of black locust (*Robinia pseudoacacia* L.) short rotation plantations in Germany: growth
727 and economics of different clones and management practices. *Bioresource Technology*, 98(2):281–286,
728 2007. ISSN 0960-8524. doi: 10.1016/j.biortech.2006.01.004.
- 729 Georg Huber, Kurt-Jürgen Hülsbergen, and Harald Schmid. Allometric equations for above-ground tree-
730 biomass estimation in Southern German short-rotation agroforestry systems. *Bioenergy Research*, 9
731 (3):955–968, 2016. ISSN 1939-1234. doi: 10.1007/s12155-016-9744-z.
- 732 Georg Huber, Christian Amann, Kurt-Jürgen Hülsbergen, and Harald Schmid. Yield potential of tree
733 species in organic and conventional short-rotation agroforestry systems in Southern Germany. *Heliyon*,
734 4(1):e00543, 2018. ISSN 2405-8440. doi: 10.1016/j.heliyon.2018.e00543.
- 735 Kaushalendra Kumar Jha. Biomass production and carbon balance in two hybrid poplar (*populus*
736 *euramericana*) plantations raised with and without agriculture in southern France. *Journal of Forestry*
737 *Research*, 29(6):1689–1701, 2018. ISSN 1007-662X. doi: 10.1007/s11676-017-0555-3.
- 738 Patrick Lamers, Martin Junginger, Carlo Hamelinck, and André Faaij. Developments in liquid biofuels
739 commodity trade. *Biofuels, Bioproducts and Biorefining*, 5(4):403–437, 2011. ISSN 1932-104X. doi:
740 10.1002/bbb.303.
- 741 Frank Lehmkuhl, Alexander Stauch, Roman Schneider, Franziska Feil, and Henning Grunewald. Short-
742 rotation coppice with poplar and willow in Germany: Biomass potentials, rotation lengths, and land-
743 use interactions. *Biomass and Bioenergy*, 176:106908, 2023. ISSN 0961-9534. doi: 10.1016/j.biombioe.
744 2023.106908.
- 745 Krishna Teja Malladi and Taraneh Sowlati. Biomass logistics: A review of important features, optimiza-
746 tion modelling and the new trends. *Renewable and Sustainable Energy Reviews*, 94:587–599, 2018.
747 ISSN 1364-0321. doi: 10.1016/j.rser.2018.06.052.
- 748 Babak Mansoornejad, Efstratios N. Pistikopoulos, and Paul R. Stuart. Integrating forest biorefinery
749 metrics into supply chain design and analysis. *Forest Policy and Economics*, 26:43–54, 2013. ISSN

1389-9341. doi: 10.1016/j.forpol.2012.08.003.

Mercer Stendal GmbH. Mercer stendal: Zellstoffwerk Arneburg – kapazitäten und rohstoffversorgung. Unternehmenswebsite, 2024. URL <https://www.mercerint.com/de/stendal/>. Abgerufen: April 2026.

P. K. Ramachandran Nair. *An Introduction to Agroforestry*. Kluwer Academic Publishers, Dordrecht, 1993. ISBN 978-0-7923-2134-7.

Marcin Niemczyk and Brian R. Thomas. Does intensive management affect wood density and fiber properties of hybrid poplar grown in short-rotation plantation? *Forests*, 12(4):399, 2021. ISSN 1999-4907. doi: 10.3390/f12040399.

Bhavna Sharma, Robert G. Ingalls, Christopher L. Jones, and Ajay Khanchi. Biomass supply chain design and analysis: Basis, overview, modeling, challenges, and future. *Renewable and Sustainable Energy Reviews*, 24:608–627, 2013. ISSN 1364-0321. doi: 10.1016/j.rser.2013.03.049.

Raffaele Spinelli, Carla Nati, and Natascia Magagnotti. Using modified foragers to harvest short-rotation poplar plantations. *Biomass and Bioenergy*, 33(5):817–821, 2009. ISSN 0961-9534. doi: 10.1016/j.biombioe.2009.01.002.

Statistisches Bundesamt (Destatis). Kaufwerte und pachtpreise für landwirtschaftliche grundstücke 2022 – Fachserie 3 Reihe 2.4. Technical report, Statistisches Bundesamt, Wiesbaden, 2023. URL <https://www.destatis.de/DE/Themen/Branchen-Unternehmen/Landwirtschaft-Forstwirtschaft-Fischerei/Flaechennutzung/Publicationen/Downloads-Flaechennutzung/kaufwerte-landwirtschaft-2030240227004.pdf>. Pachtpreise Sachsen-Anhalt, Tabelle 3.1.

Riccardo Testa, Anna Maria Di Trapani, Mario Foderà, Filippo Sgroi, and Salvatore Tudisca. Economic evaluation of introduction of poplar as biomass crop in Italy. *Renewable and Sustainable Energy Reviews*, 38:498–506, 2014. ISSN 1364-0321. doi: 10.1016/j.rser.2014.07.007.

Eric Toensmeier. The carbon farming solution: A global toolkit of perennial crops and regenerative agriculture practices for climate change mitigation and food security. In *The Carbon Farming Solution*. Chelsea Green Publishing, White River Junction, VT, 2017. ISBN 978-1-60358-571-2.

Miroslav Trnka, Jana Fialová, Vladimír Koutecký, Milan Fajman, Zdeněk Žalud, and Milan Havlíček. Biomass production and survival rates of selected poplar clones grown under a short-rotation system on agricultural land. *Plant, Soil and Environment*, 54(2):78–88, 2008. ISSN 1214-1178. doi: 10.17221/2752-PSE.

UPM Biochemicals. UPM biochemicals leuna: Bioraffinerie für laubholz-biomassekonversion. Unternehmenswebsite, 2024. URL <https://www.upmbiochemicals.com/de/>. Abgerufen: April 2026.

Dajun Yue, Fengqi You, and Seth W. Snyder. Biomass-to-bioenergy and biofuel supply chain optimization: Overview, key issues and challenges. *Computers & Chemical Engineering*, 66:36–56, 2014. ISSN 0098-1354. doi: 10.1016/j.compchemeng.2013.11.016.

Sayed Mojib Zahraee, Nirajan Shiwakoti, and Peter Stasinopoulos. Biomass supply chain environmental, economic, and social sustainability: A systematic literature review. *Industrial Crops and Products*, 145:112144, 2020. ISSN 0926-6690. doi: 10.1016/j.indcrop.2020.112144.

Table 3: Notation for the MILP model.

Sets & Indices	
$i \in \mathcal{I}$	Agroforestry sites
$j \in \mathcal{J}$	Storage / pre-treatment facilities
$k \in \mathcal{K}$	Consumer sites
$p \in \mathcal{P}$	Product types (stem, branches, residues)
$(p, p') \in \mathcal{Q}$	Admissible product-substitution pairs (quality cascade)
$\mathcal{T} = \{1, \dots, T_{\max}\}$	Planning periods (years)
$\mathcal{T}^+ = \{0, \dots, T_{\max} + 1\}$	Extended time set incl. dummy nodes
$\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$	Feasible harvest periods
$\mathcal{S}^{\text{est}} = \{(0, t) \mid t \in \{1, \dots, T_{\max} - A^{\min}\}\}$	Establishment arcs: dummy source to period t
$\mathcal{S}^{\text{harv}} = \{(s, t) \mid s, t \in \mathcal{T}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}$	Harvest cycles within age-lag bounds
$\mathcal{S}^{\text{term}} = \{(s, T_{\max} + 1) \mid s \in \{A^{\min} + 1, \dots, T_{\max}\}\}$	Termination arcs to dummy sink $T_{\max} + 1$
$\mathcal{S} = \mathcal{S}^{\text{est}} \cup \mathcal{S}^{\text{harv}} \cup \mathcal{S}^{\text{term}}$	Age-lag arc set (establishment $\cup$ harvest $\cup$ termination)
Decision Variables	
$z_{ist} \in \{0, 1\}$	1 if s and t are consecutive harvest cycles at site i
$Y_{ipt} \geq 0$	Harvest quantity (t) of product p at site i in period t
$X_{ijpt} \geq 0$	Transport flow (t) of p from site i to facility j in t
$X_{jkpp't} \geq 0$	Flow of p from facility j to consumer k satisfying demand p' in t
$S_{jpt} \geq 0$	Inventory (t) of product p at facility j end of period t
Parameters	
$T_{\max}$	Length of planning horizon (years)
$A^{\min}, A^{\max}$	Minimum / maximum stand age at harvest (years)
AREA_i	Area of site i (ha)
η_{pa}	Yield coefficient (t/ha) of product p at stand age a , derived from (??)–(??)
R_{kp}	Revenue (€/t) for product p at consumer k
C_i^{est}	Establishment cost (€/ha) at site i
C_i^{opp}	Annual opportunity cost (€/ha) at site i
C_i^{harv}	Harvest & refitting cost (€/ha) at site i
$c^{\text{tr-raw}}$	Transport cost rate (€/t km) for raw biomass
$c^{\text{tr-pre}}$	Transport cost rate (€/t km) for pre-treated biomass
c_j^{stor}	Storage holding cost (€/t) at facility j
d_{ij}, d_{jk}	Distance (km) between nodes
$\text{CAP}_j^{\text{stor}}$	Storage capacity (t) at facility j
$\text{CAP}_j^{\text{proc}}$	Processing throughput capacity (t/period) at facility j
$D_{kpt}^{\max}$	Maximum demand (t) of product p at consumer k in period t

Table 4: Consumer demand quantities and gate prices by product grade. P1: chemical/pulp; P2: sawlog/pulpwood; P3: energy/pellets. Demands in kt wood input per year; prices in €/t wood input at the factory gate.

Consumer	Annual demand			Gate price		
	P1 (kt/yr)	P2 (kt/yr)	P3 (kt/yr)	P1 (€/t)	P2 (€/t)	P3 (€/t)
consumer 1	300.0	–	–	100	–	–
consumer 2	100.0	–	–	120	–	–
consumer 3	–	300	50.0	–	75	40
consumer 4	200.0	–	–	95	–	–
consumer 5	–	30	5.0	–	70	35
consumer 6	–	15	5.0	–	65	35
consumer 7	–	10	3.0	–	60	30
consumer 8	–	5	8.0	–	55	30
consumer 9	2.0	50	20.0	80	70	38
consumer 10	–	–	1.7	–	–	35
consumer 11	–	–	2.9	–	–	35
consumer 12	–	–	2.5	–	–	35
consumer 13	–	–	2.3	–	–	35
consumer 14	–	–	1.9	–	–	35
consumer 15	–	–	2.7	–	–	35
consumer 16	–	–	1.7	–	–	35
consumer 17	–	–	1.9	–	–	35
consumer 18	–	–	1.6	–	–	35
consumer 19	–	–	1.5	–	–	35
consumer 20	–	–	1.5	–	–	35
consumer 21	–	–	1.8	–	–	35
consumer 22	–	–	3.5	–	–	35
consumer 23	–	–	1.6	–	–	35
consumer 24	–	–	2.1	–	–	35
consumer 25	–	–	4.5	–	–	35
consumer 26	–	–	1.8	–	–	35
consumer 27	–	–	3.2	–	–	35
consumer 28	18.5	–	–	55	–	–

Table 5: AFS-SCD base-case parameter values for the Saxony-Anhalt case study. Transport costs refer to road distance (OSRM). Gate prices are weighted averages across consumer sites.

Parameter	Description	Base-case value	Unit
Index sets			
$ \mathcal{T} $	Planning periods	15	yr
$ \mathcal{I} $	KUP super-sites (post-clustering)	K (clustering-dependent)	–
$ \mathcal{J} $	Pre-treatment / storage nodes	11	–
$ \mathcal{K} $	Consumer sites	9	–
$ \mathcal{P} $	Product grades	3	–
Biomass dynamics			
$a^{\min}$	Minimum harvest age (years)	4	yr
$a^{\max}$	Maximum stand age (years)	15	yr
Cost parameters			
c^{tr-raw}	Raw transport cost	0.08	€ t ⁻¹ km ⁻¹
c^{tr-pre}	Pre-processed transport cost	0.05	€ t ⁻¹ km ⁻¹
c^{opp}	Opportunity cost (foregone arable rent)	280	€ ha ⁻¹ yr ⁻¹
Revenue parameters			
$\bar{R}_{P1}$	Mean gate price P1	99	€ t ⁻¹
$\bar{R}_{P2}$	Mean gate price P2	69	€ t ⁻¹
$\bar{R}_{P3}$	Mean gate price P3	36	€ t ⁻¹